

ICT-Based Research, Innovation, and Scientific Paper Writing for Professional Engineers

BY

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Workshop Day -1

Why we are here?

By the end of the workshop, participants will be able to:

- Identify ICT research problems and innovation opportunities.
- Conduct effective literature reviews.
- Design appropriate research methodologies.
- Prepare academic and grant-oriented research proposals.
- Present and defend research ideas professionally.
- Apply innovation-driven approaches in ICT research projects.

Agenda

Research/Scientific Research

Innovation/ICT Based Innovation

Innovation Ecosystem and Digital Transformation

Research Area and Topic Selection

Group Activity: Research Problem Identification

SUMMER

WHAT MOST PEOPLE THINK
ACADEMICS DO ALL SUMMER:



YAY!
NO WORK!

WHAT MOST ACADEMICS
ACTUALLY DO:



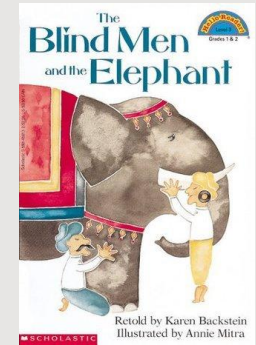
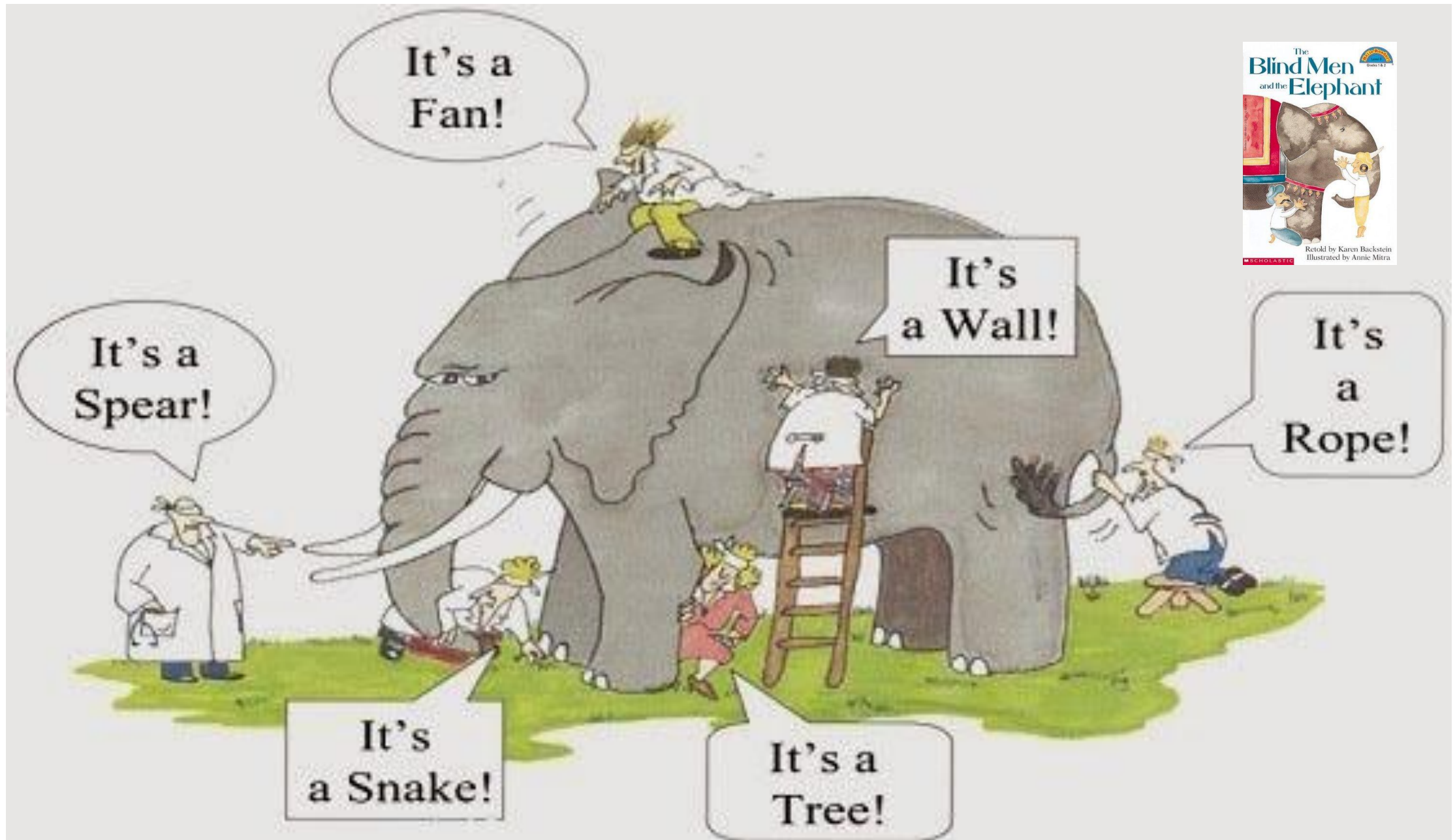
YAY! I CAN
FINALLY WORK!

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What is Research?





Research is not

Accidental discovery:

1. Accidental discovery may occur in structured research process
2. Usually takes the form of a phenomenon not previously noticed
3. May lead to a structured research process to verify or understand the observation

Research is not ...

Data Collection

- an intermediate step to gain reliable knowledge
- collecting reliable data is part of the research process

Research is not ...

Searching out published research results in libraries (or the internet)

- This is an important early step of research
- The research process always includes synthesis and analysis
- But, just reviewing of literature is not research

Research Defined and Described

“Research is the systematic approach to obtaining and confirming new and reliable knowledge”

- Systematic and orderly (following a series of steps)
- Purpose is new knowledge, which must be reliable

This is a general definition which applies to all disciplines

Research is

Searching for explanation of events, phenomena, relationships and causes

- What, how and why things occur
- Are there interactions?

A process

- Planned and managed – to make the information generated credible
- The process is creative
- It is circular – always leads to more questions

Research is

All well designed and conducted research **has potential application**.

Failure to see applications can be due to:

- Users not trained or experienced in the **specialized methods** of economic research and reasoning
- Researchers often do **not provide adequate interpretations and guidance** on applications of the research

Researchers are responsible to help users understand research implications (How?)

What is the Scientific Method?

A **systematic process** for understanding natural phenomena

- Helps **minimize bias and improve reliability**
- Builds **accurate and testable knowledge**

The Scientific Method

Observe and question

Propose tentative hypotheses

- make predictions about unobserved phenomena

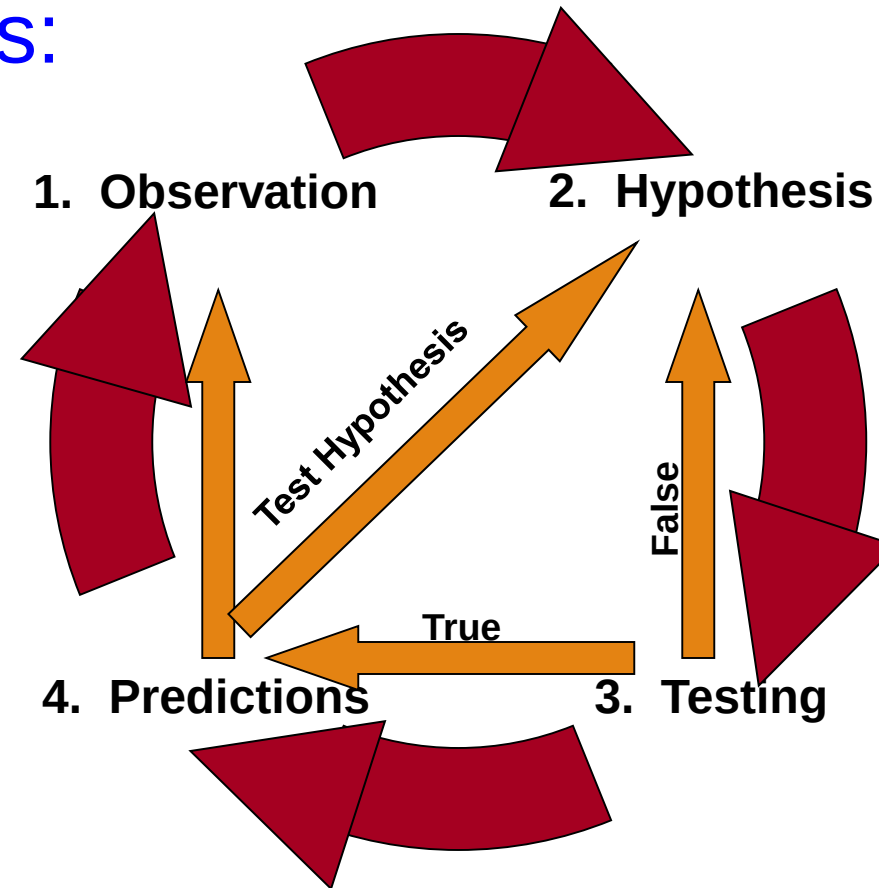
Test by making observations

Reject hypotheses that fail to predict new observations



Scientific Method of Research

Four Steps:



If the experiments prove the **hypothesis** to be true, it becomes a **theory** or **law** of nature. If the experiments prove the hypothesis to be false, the hypothesis must be rejected or modified. The scientific method used properly should give us **predictive power** (to understand phenomena which have not been tested).

Inductive
Reasoning

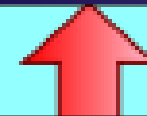
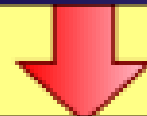
Deductive
Reasoning

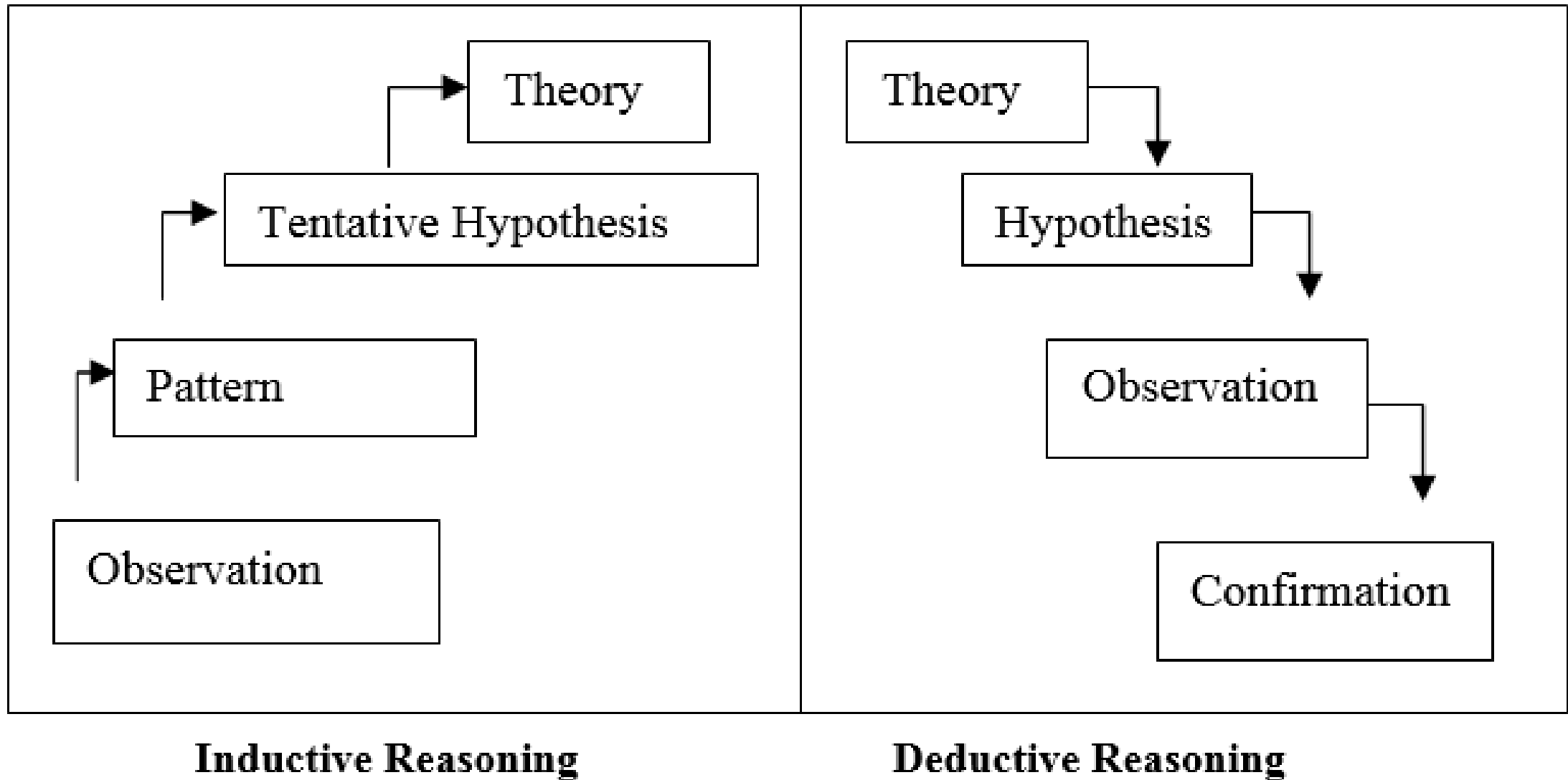
OBSERVATION / EXPERIMENT

Generalizations

Predictions

PARADIGM / THEORY





Creativity in the Research Process

Research is a creative process

“...research includes far more than mere logic ... It includes insight, genius, groping, pondering – “sense” ...
the logic we can teach; the art we cannot”

Research requires (or at least works best) with imagination, initiative, intuition, and curiosity.

There are different types of creativity, characteristic of different situations – “**applied**” and “**theoretical**” most closely associate with research

THE CREATIVE PROCESS



Fostering Creativity (Ladd 1987)

- A. Gather and use previously developed *knowledge*
- B. Exchange ideas
- C. Apply deductive logic
- D. Look at things alternate ways
- E. Question or challenge assumptions
- F. Search for patterns or relationships
- G. Take risks
- H. Cultivate tolerance for uncertainty
- I. Allow curiosity to grow
- J. Set problems aside ... and come back to them
- K. Write down your thoughts
“... frequently I don't know what I think until I write it”
- L. Freedom from distraction ... some time to think.

Creativity may provide the difference between satisfactory and outstanding research.



Master's Vs. Bachelor's Degree

The hardest part of research: asking questions

If you are smart enough to get into graduate school,

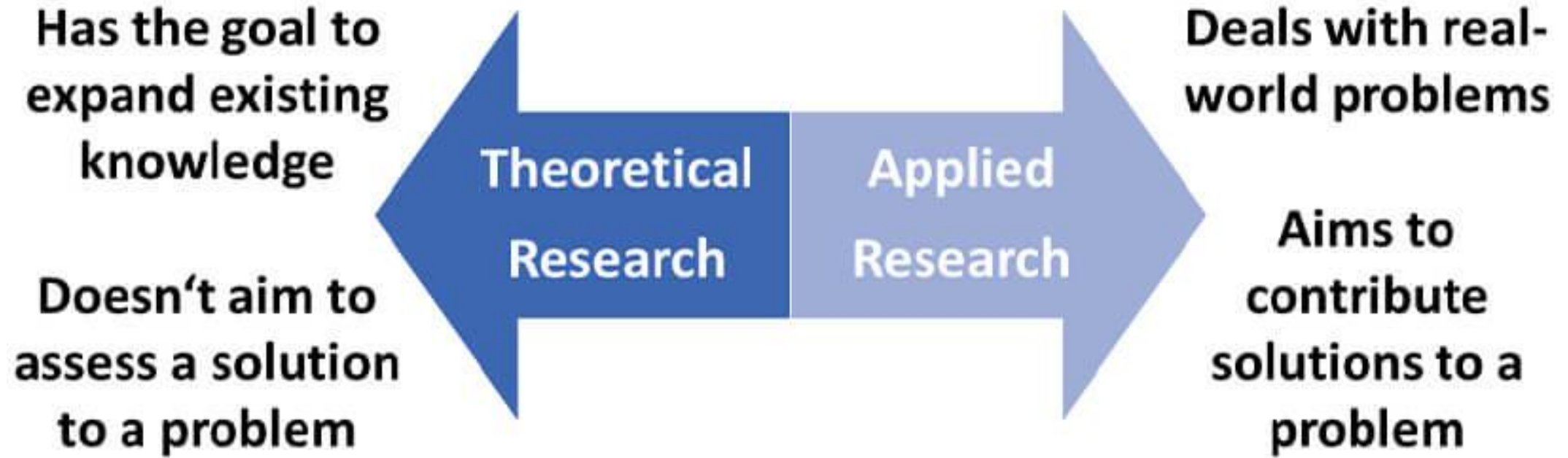
- you are probably good at **answering** questions
- That's what college trained you to learn

Graduate school teaches you how to **ask** questions

Once you ask the right question,

- the answer is often obvious

The question you ask implies a framing of the problem



Why Research?

Curiosity



Amount of
knowledge

Funding



Quality of
Life

Utility of
Applications

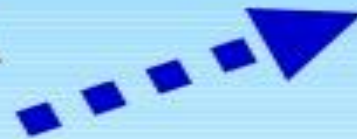
Advancement of
Technology



Basic Research



Applied Research



Application
Development



The Process of Research

- The process is initiated with a question or problem (**step 1**)
- Next, goals and objectives are formulated to deal with the question or problem (**step 2**)
- Then the research design is developed to achieve the objectives (**step 3**)
- Results are generated by conducting the research (**step 4**)
- Interpretation and analysis of results follow (**step 5**)

Common Mistakes in Research

Ignoring contradictory evidence

Bias in interpretation of data

Failure to estimate experimental errors

Scientific Research and Innovation

Importance of Scientific Research Method

- Promotes *critical thinking and innovation*
- Supports evidence-based decision making
- Widely applied in science, engineering, health, and technology

applying problem-solving skills and scientific knowledge to real-world innovation challenges

Innovation ?

“The implementation of a new or significantly improved **product** (good or service), or **process**, a **new marketing method**, or a **new organizational method** in business practices, workplace organization or external relations”. (OECD, 2018)

Mechanisms of innovation

- ✓ Novelty in **product or service** (offering something no one else does)
- ✓ Novelty in **process** (offering it in a new way)
- ✓ Complexity (offer something which others find difficult to master)
- ✓ Timing (first mover advantage, fast follower)
- ✓ Add/extend competitive factors (e.g. From price to quality or choice)
- ✓ Robust design (contribute a platform on which other variations can build)
- ✓ Reconfiguring the parts (building more effective business networks)
- ✓ ...



Linear View of Innovation

(Traditional Approach)

Innovation was initially seen as a **linear process**

Sequence: **Science → Technology → Products/Services → Market needs**

Assumption: More **R&D investment = more innovation**

Focus mainly on **inputs like research and funding**

Limitation:

- Ignores real-world complexity of innovation
- Oversimplifies how innovation actually happens

Systems View of Innovation

(Modern Approach)

Innovation is not a Linear Process

➤ Innovation is a **complex, interactive system**

➤ Driven by multiple interconnected factors:

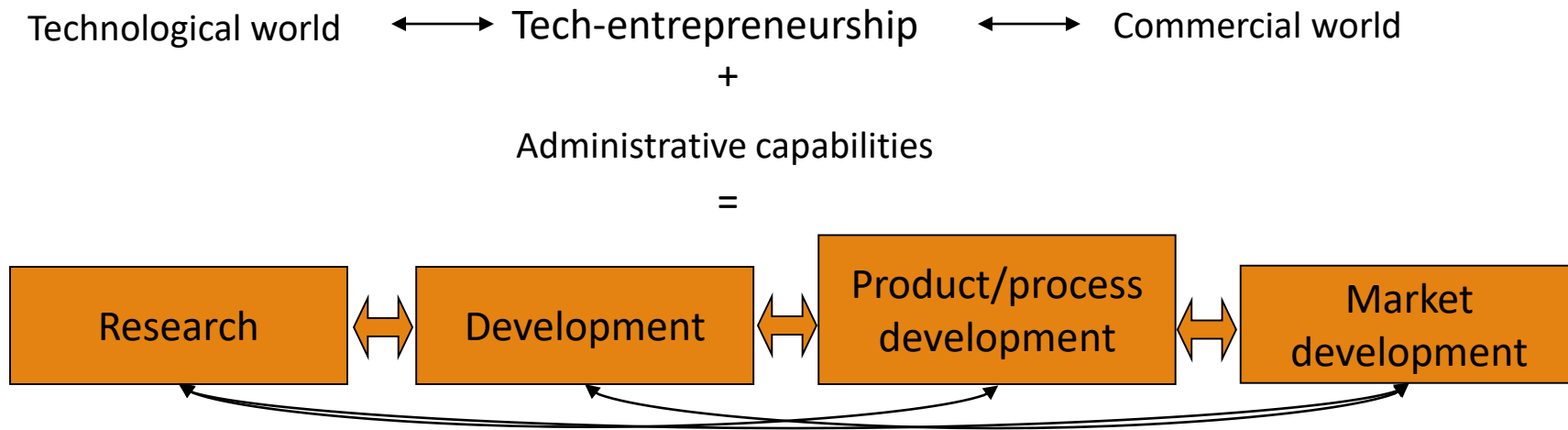
- R&D and investment
- Education and talent
- Markets and economic conditions
- Culture and networking
- Institutions and policy
- Serendipity and collaboration

➤ Emphasis on **interaction among stakeholders**

➤ Innovation emerges from links between:

- Science, technology, society, and policy
- Success depends on understanding:
 - Roles of actors
 - Relationships within the system
 - Coordinated ecosystem development

Today's basic model for innovation management is interactive



ICT-Centric Innovation

A description of an innovation ecosystem recognizing that ICTs are often at the centre of innovation, and have a cross cutting role in many other sectors of the economy.

ICT-centric innovation has two main characteristics.

Development of the ICT sector



Cross-cutting role of ICT innovation



An ecosystem and its stakeholders

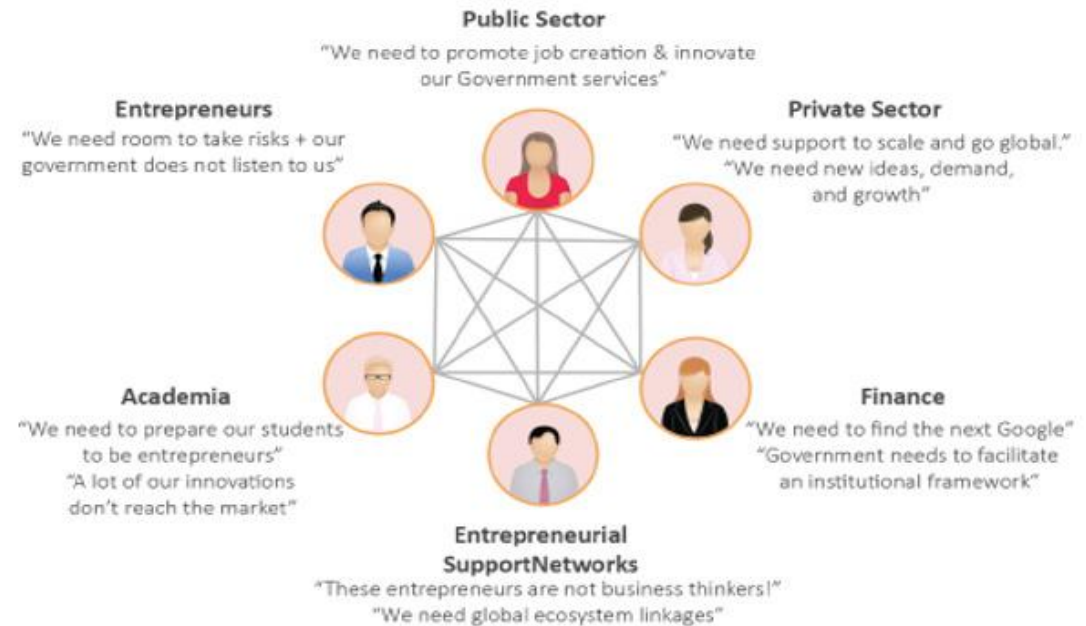
ITU defines an ecosystem as:

*“A system or network of interconnecting and interacting organizations and stakeholders, from multiple sectors, who **come together and address the problems** people are facing within their communities”.*

What happens in an ecosystem if its stakeholders do not interact?

- ✓ The short answer is that it will **struggle to be sustainable.**

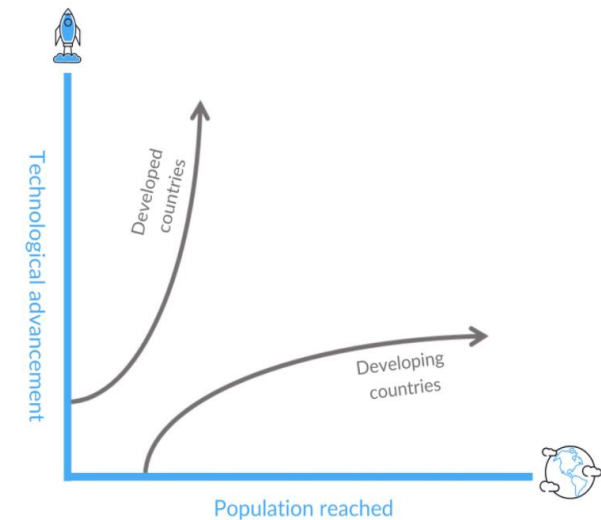
An ICT-centric ecosystem will be sustainable when everyone understands their role and collaborates with each other to create impact.



Rethinking the digital divide

- Expanding on a basic understanding of innovation systems, the focus is now on the **innovation divide—a significant barrier to the pursuit of inclusive economic growth.**
- Indeed, **not all innovation ecosystems are created equal.**
 - There are vast differences between ecosystems both in terms of **available resources** and **how those resources are applied**, and the **enabling environment**, which can **lead to very different outcomes.**
- **The creation and benefits of technology are currently concentrated in a small number of countries and communities.**

Technological advancement vs. population reached:
a tale of innovation in developed and developing countries



Source: ITU [Bridging the digital innovation divide: A toolkit for strengthening ICT-centric innovation ecosystems](#)

Innovation divide

Rapid technological change creates **uneven opportunities between developed and developing countries** due to differences in innovation capacity.

The **digital divide reflects unequal ability to adopt and use technological innovation**, leading to gaps in economic and innovation development.

The root cause of the divide is **low innovation capability and weak collaboration among stakeholders**, along with limited access to appropriate policies, programs, and resource

"This is a new era of opportunity, but only for those who are willing to accept change as an opportunity, not for those who are afraid of it".

Peter Drucker

Innovation in service: Airbnb

Founded: 2008 — Digital platform for home-sharing & travel services

Market Position: ~8 million listings across 220+ countries/regions

Industry Scale: Estimated market value over \$80 billion in hospitality services

Key Innovations: Mobile booking, AI recommendations, instant booking, digital payments

Service Expansion: Airbnb Experiences added local tours & activities

Industry Impact: Disrupted hotel industry and accelerated the sharing economy model

Customer Reach: 1.5+ billion guest arrivals globally since launch

Business Impact: Created income opportunities for millions of hosts worldwide

Airbnb reshaped hospitality through digital innovation and platform-based services.

<https://youtu.be/16cM-RFid9U>

Discussions:

Major Intervention based on Research Findings

Sources of research ideas

- Always be working on *something* (even if it won't be your thesis topic)
 - Think about problems, difficulties, or irritations
 - You might get sidetracked onto something more important
- Learn about other research: attend talks, read papers, take classes
 - future work they identified
 - connections they did not make
 - assumptions
 - other criticisms
- Your advisor
 - OK for starter projects, master's projects
 - not OK for research/thesis topic, specifically for MPhil/PhD



Sources of research ideas

Think outside the box: A good research idea often “cheats”

- If you take the same approach as other people,
you are likely to be no more successful than they are
- **Cut the Gordian knot:** take a different approach



Bad ideas are essential

- Most ideas are bad ideas
- If you have lots of ideas, some will be good
- It's better to have bad ideas than no ideas
 - I can teach people how to filter
 - It's harder to teach courage and imagination
- Filtering is initially your advisor's role
 - You will get better with time, and spend less time evaluating and discarding bad ideas
 - Have faith in your own judgment too
 - Your advisor is not infallible
- Generate lots of ideas before filtering
 - **Breadth-first, not depth-first, approach to the problem**

Research Area/Topic Selection

A well-selected topic is the foundation of successful research.

A strong research area leads to meaningful and sustainable research.

Before Selecting a Research Area Consider

- Personal interest
- Academic background
- Available resources
- Research scope
- Industry relevance
- Supervisor expertise
- Future career opportunities

Purpose of the
Research ?

Important: Choose an area you can continue exploring for years.

Who are you ?

Ask Yourself:

- Which subjects do I enjoy most?
- Which technologies excite me?
- What problems do I want to solve?
- Which courses/projects interested me most?

STRENGTHS	WEAKNESSES
What research skills do I already have? Which subjects or technologies am I most confident in? What programming, analytical, or laboratory skills can support my research? What previous projects or coursework relate to this topic? Do I have strong interest and motivation in this area? What advantages do I have compared to others working in similar areas? Do I have access to supervisors, labs, or institutional support?	Which technical skills do I lack for this research area? Do I have limited knowledge of research methodology or tools? Do I lack access to datasets, equipment, or software? Is the research area too broad or too difficult for me? Do I have time limitations or resource constraints? What areas require additional training or guidance? Am I unfamiliar with recent literature or trends?
OPPORTUNITIES	THREATS
What current research trends make this area valuable? Are there real-world or industry problems that need solutions? Are funding, grants, or collaboration opportunities available? Can this research lead to publication or innovation? Is there increasing demand for experts in this research area? Can interdisciplinary approaches create new research opportunities? Does the topic have long-term relevance and impact?	Are there rapidly changing technologies that may make the topic outdated? Is the research area highly competitive or saturated? Are there limitations in data availability or ethical approval? Could lack of infrastructure or computational resources affect progress? Are there external risks such as policy, technology, or market changes? Could dependency on expensive tools/software hinder the research? Are there anticipated challenges in implementation or experimentation?



Selecting a Research Area

1. Literature Review

- Read recent IEEE/ACM/Springer papers
- Identify emerging technologies and research gaps

2. Problem-Based Approach

- Focus on real-world challenges
- Example: IoT security, smart healthcare

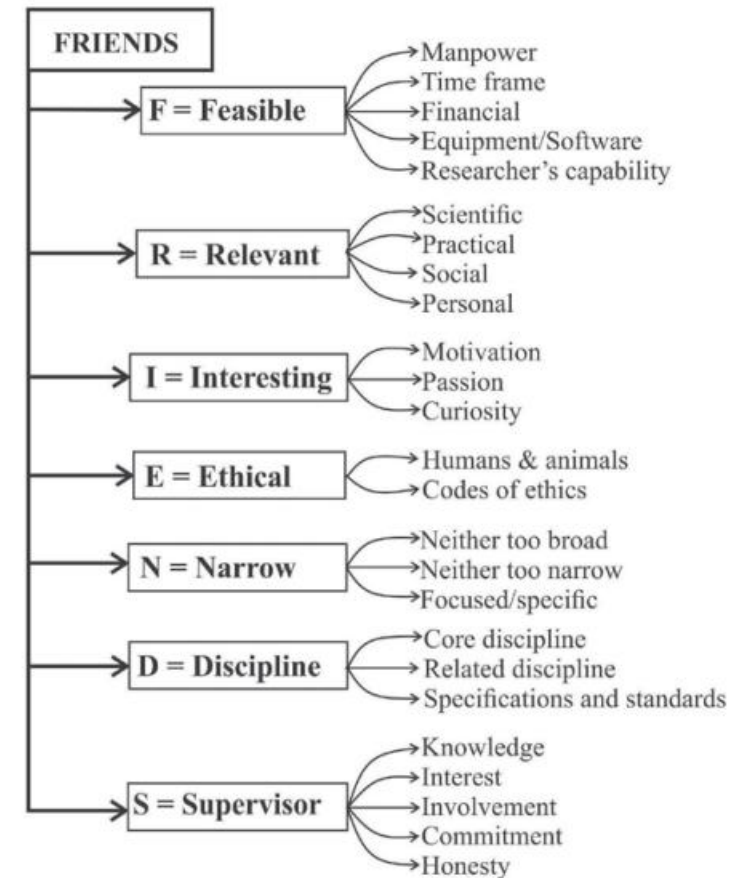
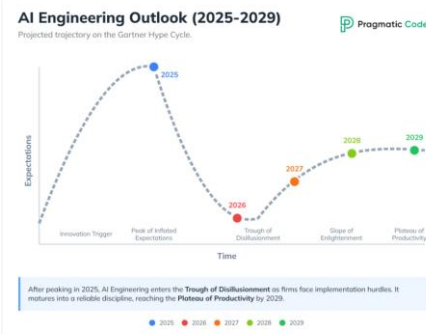
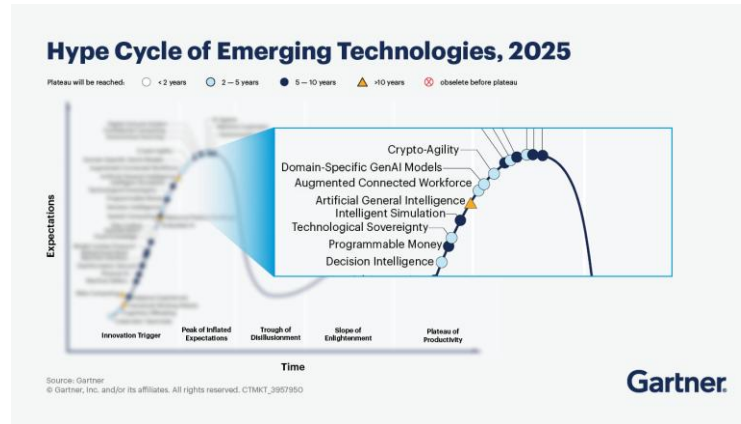
3. Trend Analysis

- Current popular areas:
- Artificial Intelligence
- Cybersecurity
- IoT
- Cloud & Edge Computing
- Blockchain
- 5G/6G Networks



4. Narrowing the Scope

- Broad Area → Specific Area → Research Problem
- Computer Networks → IoT Networks → IoT Security → Intrusion Detection



Framework for topic selection (Adhikari)